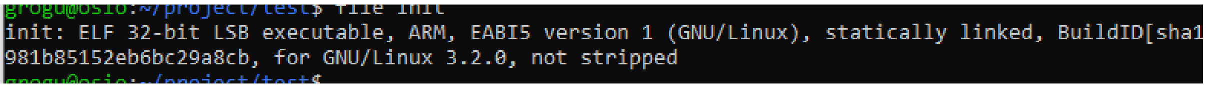
Ядро после запуска ищет программу init. Она хранится в корневой файловой системе.

В лабораторной работе надо написать свой init процесс.

1) Реализация процесса init



```
init: ELF 32-bit LSB executable, ARM, EABI5 version 1 (GNU/Linux), statically linked, BuildID[sha1
981b85152eb6bc29a8cb, for GNU/Linux 3.2.0, not stripped
```

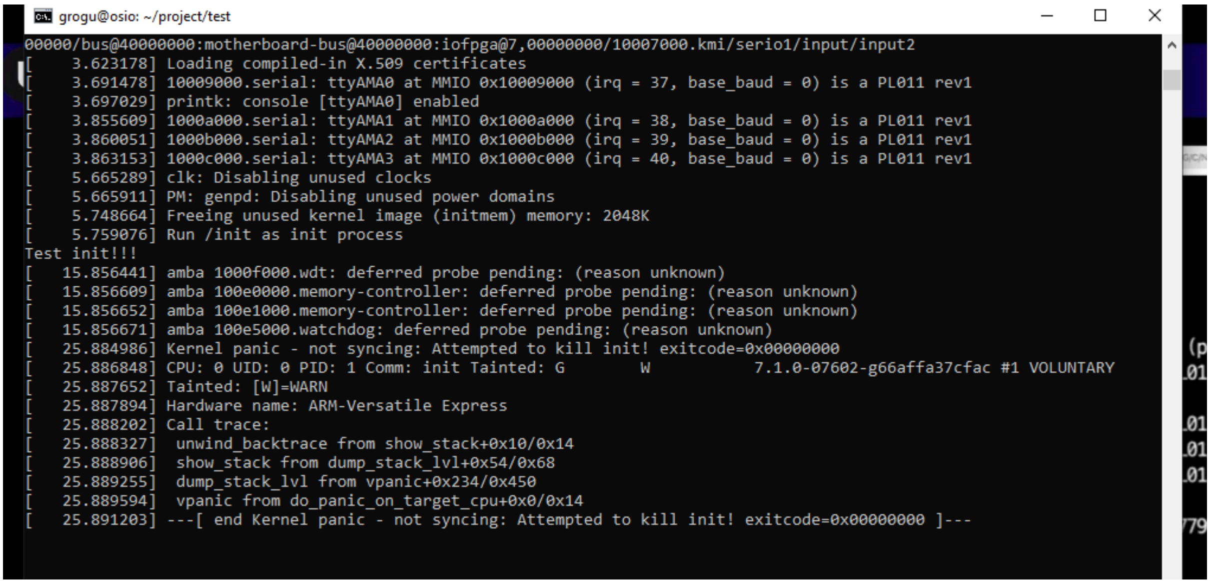
```
#include <stdio.h>
#include <unistd.h>

int main(){
	printf("Test init!!!\n");
	sleep(20);
}
```

2) создание минимального initramfs (начальный RAM-диск), который содержит только один файл — исполняемый файл.

   echo init | cpio -o -H newc | gzip > initramfs.cpio.gz

3) Запуск эмулятора
   QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -kernel zImage -initrd initramfs.cpio.gz -dtb vexpress-v2p-ca9.dtb -append "console=ttyAMA0" -nographic

4) Выполнение бинарника. Вывело Test init!!! и ядро снова упало.



```
grogu@osio: ~/project/test
00000/bus@40000000:motherboard-bus@40000000:iofpga@7,00000000/10007000.kmi/serio1/input/input2
[    3.623178] Loading compiled-in X.509 certificates
[    3.691478] 10009000.serial: ttyAMA0 at MMIO 0x10009000 (irq = 37, base_baud = 0) is a PL011 rev1
[    3.697029] printk: console [ttyAMA0] enabled
[    3.855609] 1000a000.serial: ttyAMA1 at MMIO 0x1000a000 (irq = 38, base_baud = 0) is a PL011 rev1
[    3.860051] 1000b000.serial: ttyAMA2 at MMIO 0x1000b000 (irq = 39, base_baud = 0) is a PL011 rev1
[    3.863153] 1000c000.serial: ttyAMA3 at MMIO 0x1000c000 (irq = 40, base_baud = 0) is a PL011 rev1
[    5.665289] clk: Disabling unused clocks
[    5.665911] PM: genpd: Disabling unused power domains
[    5.748664] Freeing unused kernel image (initmem) memory: 2048K
[    5.759076] Run /init as init process
Test init!!!
[   15.856441] amba 1000f000.wdt: deferred probe pending: (reason unknown)
[   15.856609] amba 100e0000.memory-controller: deferred probe pending: (reason unknown)
[   15.856652] amba 100e1000.memory-controller: deferred probe pending: (reason unknown)
[   15.856671] amba 100e5000.watchdog: deferred probe pending: (reason unknown)
[   25.884986] Kernel panic - not syncing: Attempted to kill init! exitcode=0x00000000
[   25.886848] CPU: 0 UID: 0 PID: 1 Comm: init Tainted: G        W           7.1.0-07602-g66affa37cfac #1 VOLUNTARY
[   25.887652] Tainted: [W]=WARN
[   25.887894] Hardware name: ARM-Versatile Express
[   25.888202] Call trace:
[   25.888327]  unwind_backtrace from show_stack+0x10/0x14
[   25.888906]  show_stack from dump_stack_lvl+0x54/0x68
[   25.889255]  dump_stack_lvl from vpanic+0x234/0x450
[   25.889594]  vpanic from do_panic_on_target_cpu+0x0/0x14
[   25.891203] ---[ end Kernel panic - not syncing: Attempted to kill init! exitcode=0x00000000 ]---
```

5) Дальше нужно поставить в эмулятор полноценную корневую файловую систему busybox.
   `wget https://busybox.net/downloads/busybox-1.36.1.tar.bz2`

tar -xjf busybox-1.36.1.tar.bz2

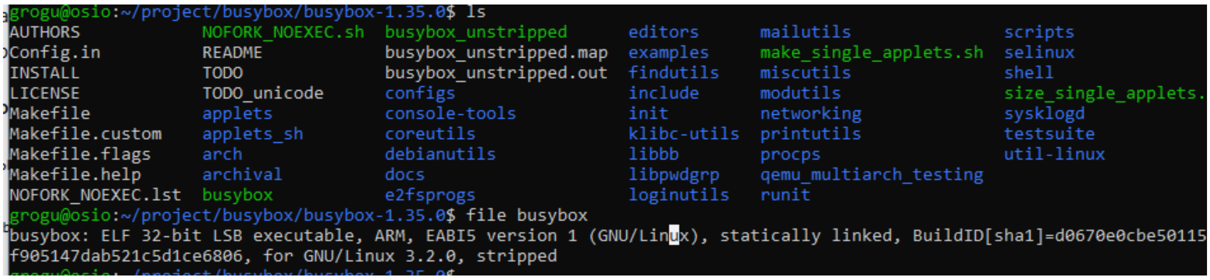
6) Сборка дефолтного конфига под arm

ARCH=arm make defconfig

7) По умолчанию busybox собирается с динамическими библиотеками. Нужно поправить конфиг, чтобы он собирался статически.

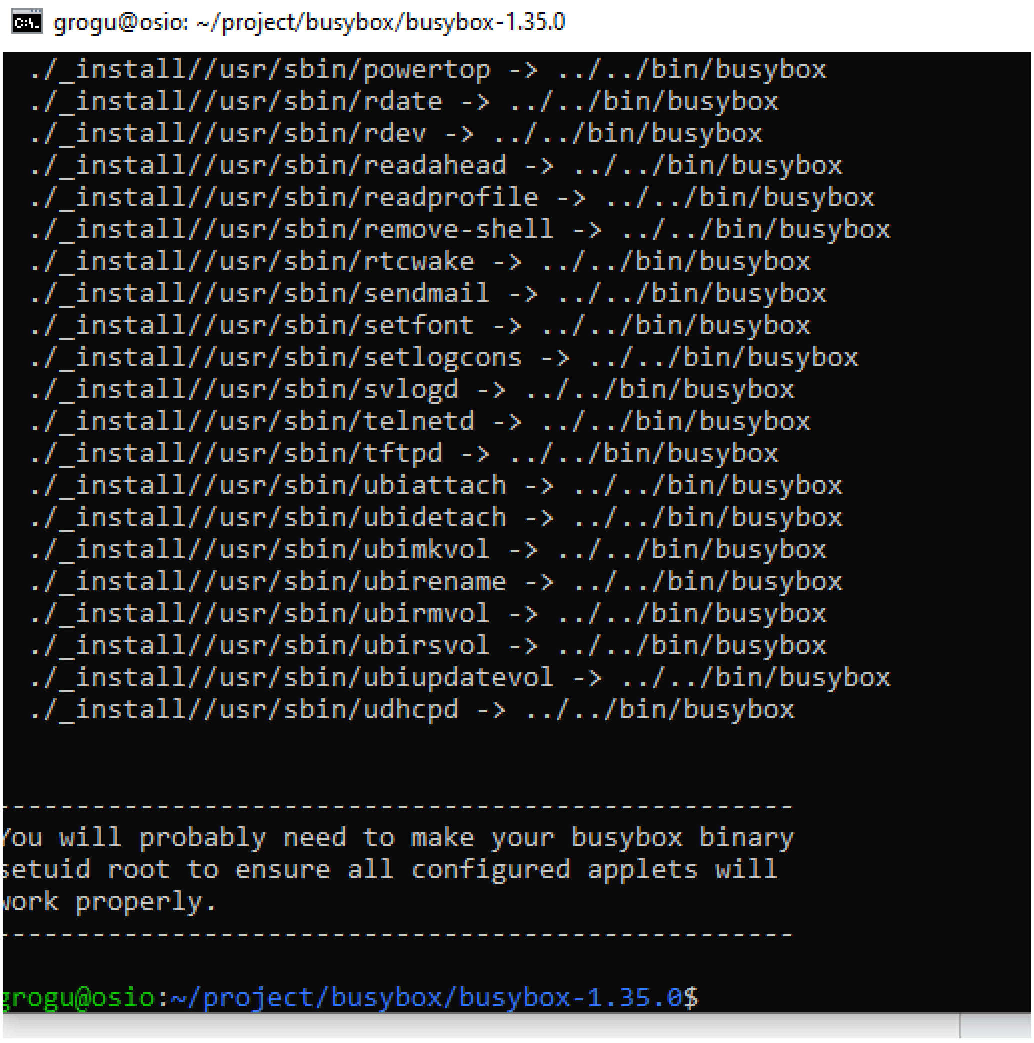
8) Собираем корневую файловую систему

make ARCH=arm CROSS_COMPILE=arm-linux-gnueabihf- -j$(nproc)



9) Устанавливаем

make install



10) Создаем архив

```
find . | cpio -o -H newc | gzip > initramfs.cpio.gz
```

11) Запускаем эмулятор

12) QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -kernel zImage -initrd initramfs.cpio.gz -dtb vexpress-v2p-ca9.dtb -append "console=ttyAMA0 rdinit=/bin/ash" -nographic

```
[    1.976900] mmci-pl18x 10005000.mmci: mmc0: PL181 manf 41 rev0 at 0x10005000 irq 33,34 (pio)
[    2.018754] 10009000.uart: ttyAMA0 at MMIO 0x10009000 (irq = 37, base_baud = 0) is a PL011 rev1
[    2.059038] printk: console [ttyAMA0] enabled
[    2.060796] 1000a000.uart: ttyAMA1 at MMIO 0x1000a000 (irq = 38, base_baud = 0) is a PL011 rev1
[    2.062090] 1000b000.uart: ttyAMA2 at MMIO 0x1000b000 (irq = 39, base_baud = 0) is a PL011 rev1
[    2.063352] 1000c000.uart: ttyAMA3 at MMIO 0x1000c000 (irq = 40, base_baud = 0) is a PL011 rev1
[    2.067193] rtc-pl031 10017000.rtc: registered as rtc0
[    2.068034] rtc-pl031 10017000.rtc: setting system clock to 2026-05-28T13:14:05 UTC (1779974045)
[    2.075248] uart-pl011 10009000.uart: no DMA platform data
[    2.103626] Freeing unused kernel memory: 2048K
[    2.105992] Run /bin/ash as init process
[    2.121146] input: AT Raw Set 2 keyboard as /devices/platform/bus@4000000/bus@4000000:motherboard/bus@4000000:motherboard:iofpga@7,0000000
/10006000.kmi/serio0/input/input0
/bin/ash: can't access tty; job control turned off
~ # [    2.764170] input: ImExPS/2 Generic Explorer Mouse as /devices/platform/bus@4000000/bus@4000000:motherboard/bus@4000000:motherboard:io
pga@7,00000000/10007000.kmi/serio1/input/input2

~ #
~ #
~ #
~ #
```